FlashInfer AI Kernel Generation Contest @ MLSys 2026

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1) Submission context

- **Track:** `fused_moe` (`moe_fp8_block_scale_ds_routing_topk8_ng8_kg4_e32_h7168_i2048`)
- **Kernel implementation:** `solution\triton\kernel.py` (run entry point)
- **Benchmark target:** NVIDIA B200 via `scripts\run_modal.py`
- **Agent stack:** GitHub Copilot CLI agents, **GPT-5.3-Codex (xhigh)**, and Andrej Karpathy's **autoresearch** agent skill
- **Method class:** Agent-Assisted, with an autoresearch-style autonomous experiment loop

2) End-to-end kernel working (full explanation)

2.1 Problem shape and fixed constants

In `run()`, the kernel is specialized for:

- `H=7168` (model hidden)
- `I=2048` (intermediate expert dimension)
- `E_global=256`, `E_local=32`
- `TOP_K=8`, `N_GROUP=8`, `TOPK_GROUP=4`
- block sizes: `BLOCK_M=16`, `BLOCK_K=128`, `BLOCK_I=128`, `BLOCK_N=128`

These choices define tile geometry and dispatch behavior across all routed workloads.

2.2 Stage A: compiled routing + dispatch (`_routing_and_dispatch`)

This stage converts routing logits into expert-token assignments and per-route weights:

1. Convert logits to FP32 and apply sigmoid.
2. Add routing bias.
3. Group experts (`N_GROUP=8`) and compute group scores from top-2 experts per group.
4. Keep top `TOPK_GROUP` groups per token.
5. Mask pruned experts and select final `TOP_K=8` experts per token.
6. Normalize selected routing weights with scaling.
7. Build dispatch order:
    - compute local expert ids
    - stable sort by local expert id
    - produce `sorted_tokens_all`, `sorted_weights_all`
8. Compute `expert_counts`, `expert_offsets`, and `block_offsets`.

This function is wrapped with: `torch.compile(_routing_and_dispatch, mode="reduce-overhead")` to reduce Python overhead and fuse routing-side ops into a lower-overhead execution path.

2.3 Stage B: GPU block map build (`_build_block_map_kernel`)

Given `block_offsets` and `expert_offsets`, this kernel builds three block metadata arrays:

- `b_expert_id`
- `b_token_offset`

- `b_num_tokens`

Each Triton program handles one output block id and:

1. finds owning expert (linear scan over `E_LOCAL=32`)
2. computes token offset for that block
3. computes valid token count (tail-aware, max `BLOCK_M`)

Why this matters: it avoids a multi-op CPU/PyTorch block-map pipeline and keeps mapping on device.

2.4 Stage C: GEMM1 + SwiGLU (`_moe_gemm1_swiglu_kernel`)

Grid: `(total_blocks, NUM_I_BLOCKS)` where:

- program axis 0 = routed block id
- program axis 1 = intermediate block index (`ib`)

For each routed block and each intermediate tile:

1. load token ids from `sorted_tokens`
2. gather FP8 hidden states + per-block hidden-state scales
3. load FP8 expert weights for W1 and W3 + scales
4. compute two FP32 dot products
5. apply dequant scaling (`sA * sW1`, `sA * sW3`)
6. apply SwiGLU (`silu(u2) * u1`)
7. store FP32 activations to `workspace[total_routed, I]`

This stage is compute-heavy and quantization-aware; FP32 accumulation is used for correctness stability.

2.5 Stage D: GEMM2 (`_moe_gemm2_kernel`)

Grid: `(NUM_H_BLOCKS * total_blocks,)` with grouped launch ordering.

Core mechanics:

1. map `pid` to `(block_id, nb)` using `GROUP_BLOCKS`
2. load routed tokens and route weights for that block
3. iterate `ib` over `NUM_I_BLOCKS`:
    - load FP32 workspace tile
    - load FP8 W2 tile + scale
    - dot in FP32 and apply W2 scale
4. multiply accumulated output by per-route weight
5. `tl.atomic_add` into global FP32 accumulator (`out_accum`)

Important correctness/performance controls:

- autotune configs over `(GROUP_BLOCKS, num_warps, num_stages)`
- autotune key `['TOTAL_BLOCKS', 'TOTAL_ROUTED']`
- `reset_to_zero=['out_ptr']` to keep atomic accumulation correct across autotune trials

2.6 Stage E: output finalize

`out_accum` is explicitly initialized with `torch.zeros((T,H), dtype=float32)` and copied to `output`. The explicit zero init is required for correctness in this path.

3) Performance evolution and bottlenecks

From `research\results.tsv` and `research\kernel_autoresearch.md`:

- **Total logged experiments:** 299
- **Status split:** 17 keep / 235 discard / 46 crash
- **Best run:** `0.995000 ms` at commit `79897a020fd59648100377d9da4c41e7413f6f33`

Key milestones:

Milestone	Avg latency (ms)	Effect
Baseline (`0cc6daf`)	1.524421	Starting point
Grouped GEMM2 ordering (`bd4cfe7`)	1.264368	Large L2 locality gain
GEMM2 autotune + reset-to-zero (`53cce00`)	1.088105	Major step down
Add `TOTAL_ROUTED` key (`3f14870`)	1.085474	Better shape-aware config pick
Add `g4,w4,s3` config	1.009474	Near-threshold
Focused rerun (`79897a...`)	**0.995000**	Correctness-clean sub-1ms

Dominant remaining issues:

- long-sequence workloads (`5e8dc11c`, `58a34f27`) remain the slowest
- heavy run-to-run variance near the 1ms boundary
- many low-precision or aggressive scheduling changes caused numerical failures

4) Andrej Karpathy's autoresearch agent skill adaptation for this project

We used Andrej Karpathy's autoresearch agent skill as the template for autonomous experiment design.

4.1 What we adopted from Andrej Karpathy's autoresearch agent skill

From the autoresearch README and `program.md` workflow pattern, we adopted:

- **single primary mutable code surface**
- **fixed, repeatable benchmark loop**
- **strict experiment ledger (`keep` / `discard` / `crash`)**
- **autonomous keep-or-revert discipline**
- **continuous agent loop behavior ("never stop" style until interrupted)**

4.2 Translation from autoresearch to kernel optimization

Original autoresearch pattern	Our MoE kernel adaptation
Mutate `train.py` only	Mutate `solution\triton\kernel.py` (+ minimal config/log updates)
Run `uv run train.py`	Run pack + Modal benchmark pipeline
Optimize `val_bpb`	Optimize average latency (ms) under correctness
Track `results.tsv` with keep/discard/crash	Same structure in `research\results.tsv`
Use `program.md` as agent skill	Use autoresearch-style instructions + contest-specific constraints for agent actions

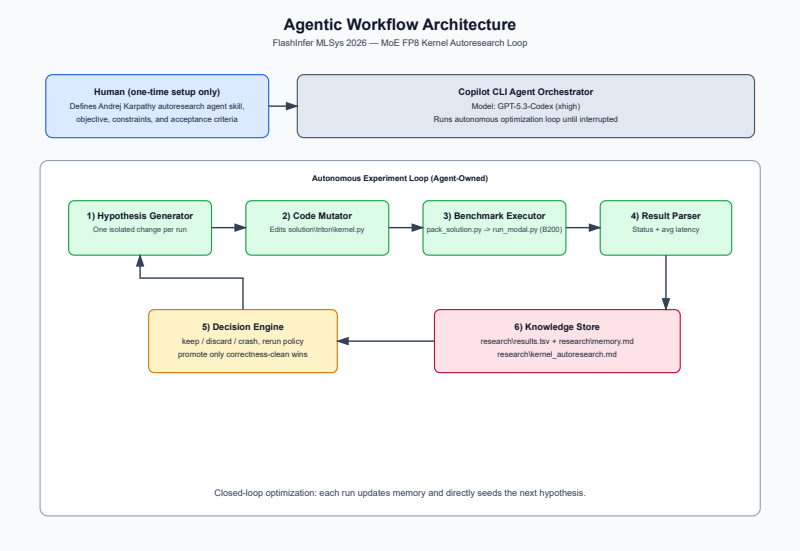
4.3 Concrete auto-research loop used here

Per iteration:

1. Agent proposes one hypothesis.
2. Agent edits kernel code.
3. Agent packs solution: `conda run --no-capture-output -n fi-bench python scripts/pack_solution.py`
4. Agent benchmarks on B200: `conda run --no-capture-output -n fi-bench modal run scripts/run_modal.py > run.log 2>&1`
5. Agent parses benchmark outcomes from `run.log`.

6. Agent logs experiment in `research\results.tsv`.
7. Agent updates research memory (`research\memory.md`, `research\kernel_autoresearch.md`).
8. Agent decides keep/revert and immediately starts next trial.

5) Workflow architecture (proper agentic architecture)



This creates a closed autonomous optimization loop where evidence from each run directly drives the next code mutation.

6) Human vs agent responsibility (as requested)

For this submission, the practical split was:

- **Human:** implemented and provided the autoresearch skill-agent setup/instructions and constraints.
- **Agent:** executed the research campaign end-to-end (idea generation, code edits, benchmark execution, result parsing, logging, keep/revert decisions, and iterative optimization).

So after setup, experiment execution was agent-driven.

7) Tools and stack

- Triton + PyTorch for kernel implementation
- FlashInfer-Bench workflow for evaluation
- Modal for B200 benchmarking
- Andrej Karpathy's autoresearch agent skill methodology for autonomous research loop structure
- GitHub Copilot CLI agents powered by **GPT-5.3-Codex (xhigh)**

8) Reproducibility

From repo root:

```
conda run --no-capture-output -n fi-bench python scripts/pack_solution.
conda run --no-capture-output -n fi-bench modal run scripts/run_modal.p
```

Then append experiment metadata to `research\results.tsv`:

- commit or run id
- avg latency (ms)
- status (`keep`, `discard`, `crash`)
- concise description of attempted change

9) Final outcome

The autoresearch-driven agent workflow reached a correctness-clean **0.995000 ms** average latency run while preserving strict numerical constraints. The best path was centered on GEMM2 grouped scheduling and focused autotune-pool refinement, with long-sequence variance still the main remaining optimization frontier.

10) References

10.1 Internal project references

- `solution\triton\kernel.py`
- `research\results.tsv`
- `research\memory.md`
- `research\kernel_autoresearch.md`
- `scripts\pack_solution.py`
- `scripts\run_modal.py`
- `README.md`
- `autoresearch\README.md`
- `autoresearch\program.md`

10.2 External references used across the project

- Contest and benchmark stack:
  - http://mlsys26.flashinfer.ai/
  - https://github.com/flashinfer-ai/flashinfer-bench-starter-kit
  - https://github.com/flashinfer-ai/flashinfer
  - https://github.com/flashinfer-ai/flashinfer-bench
  - https://huggingface.co/datasets/flashinfer-ai/mlsys26-contest
  - https://bench.flashinfer.ai/docs/flashinfer-trace
  - https://www.nvidia.com
  - https://modal.com
- Triton/kernel optimization references:
  - https://triton-lang.org/main/getting-started/tutorials/03-matrix-multiplication.html
  - https://triton-lang.org/main/getting-started/tutorials/08-grouped-gemm.html
  - https://triton-lang.org/main/getting-started/tutorials/09-persistent-matmul.html
  - https://triton-lang.org/main/python-api/generated/triton.language.range.html
  - https://triton-lang.org/main/python-api/generated/triton.language.make_tensor_descriptor.html
  - https://triton-lang.org/main/python-api/generated/triton.language.load.html
  - https://triton-lang.org/main/python-api/generated/triton.language.store.html
  - https://raw.githubusercontent.com/triton-lang/triton/main/python/tutorials/11-programmatic-dependent-launch.py
  - https://raw.githubusercontent.com/triton-lang/triton/main/python/tutorials/gluon/04-tma.py
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  - https://www.nvidia.com/en-us/data-center/gb200-nvl72/
  - https://arxiv.org/abs/2401.06066
  - https://arxiv.org/abs/2405.04434
  - https://arxiv.org/abs/2412.19437

    - https://pytorch.org/blog/training-moes/
- Autoresearch methodology references:
    - https://github.com/karpathy/autoresearch
    - https://docs.astral.sh/uv/
    - https://astral.sh/uv/install.sh
    - https://github.com/jsegov/autoresearch-win-rtx